

Oussama Abderrahmane Dairi

CS GRADUATE · SECURITY AND NETWORKS

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Summary

Systems programmer passionate about OS internals, network protocols, and low-level security. My research interests center on understanding and hardening system interfaces, from the kernel to the network stack, with a focus on exploitation techniques and defense mechanisms. Looking to apply deep systems knowledge to hard security problems.

Academic Background

Lycée Malika Gaid

BACCALAURÉAT SCIENCES NATURELLES

Setif, Algeria

Sept. 2020

École Supérieure en informatique 08 Mai 1945

CYCLE PRÉPARATOIRE

Sidi Bel Abbés, Algeria

Dec. 2020 - Jul. 2022

École Nationale Supérieure d'Informatique

MASTER AND ENGINEER DEGREE IN COMPUTER SYSTEMS

Alger, Algeria

Sep. 2022 - Sep. 2025

Top Courses : Advanced Networking, Cybersecurity, High-performance Computing, Distributed Systems

Internships

Web Development Intern

IRIS / EURL SATEREX

Setif, Algeria

Jul. 2023 - Sept. 2023

- Identified inefficiencies in the employee leave management process and proposed the development of a dedicated web-based solution.
- Designed and implemented the web application solution to manage leave requests and approval workflows with role-based access control for different organizational roles.

Research Intern

LABORATOIRE INFORMATIQUE, IMAGE ET INTERACTION (L3I)

La Rochelle, France

Sept. 2024 - Jul. 2025

- Reviewed state-of-the-art autonomous exploitation methods and identified key research gaps.
- Designed and evaluated a multi-agent framework to study protocol-aware planning and adaptive strategies.
- Developed reproducible experiments and benchmarks, analyzing metrics to draw insights on system performance.
- The internship's work resulted in a paper submitted to ACNS 2026.

Projects

INVEST

RESEARCH INTERNSHIP PROJECT

Jan. 2025 - Jul. 2025

- A two-agent LLM framework (Supervisor-Executor) to autonomously plan and execute packet-level attacks across OSI Layers 2-7.
- Implemented bounded feedback-loop control using stdout/stderr parsing and packet-level KPI validation for adaptive replanning.
- Designed reproducible benchmark of 7 network attack scenarios (flooding, ARP spoofing, TCP hijacking, CVE exploitation) in isolated VM testbeds.

Network Configuration and SIEM Integration

FOURTH YEAR ACADEMIC PROJECT

Mar. 2024 - Jul. 2024

- Designed and simulated a network infrastructure using GNS3 and VMware.
- Engineered a defensive stack by configuring a pfSense firewall, deploying a Snort IDS, and centralizing logs with a Wazuh SIEM.
- Validated the setup by simulating network attacks and successfully analyzing the generated alerts.

Development of security challenges for CTF competitions

SHELLMATES CLUB

2023 - 2024

- Authored binary exploitation and reverse engineering challenges for national CTF competitions (HackINI 2k24, DevFest 2023).

Networking and Systems Projects

PERSONAL PROJECTS

2023 - 2024

- Built a **terminal-based text editor**, demonstrating system programming and terminal interface skills.
- Developed a **TCP chat in C**, gaining hands-on experience with networking and sockets.
- Implemented a **binary loader**, deepening understanding of executable formats and their structures.

Skills

Programming and scripting, Python, Bash, C, C++, JavaScript, PHP, Java

Cybersecurity, Web security, Binary Exploitation, Reverse Engineering, SIEM, Network Security

Tools, Ghidra, GDB, Wireshark, Burp Suite, GNS3, VirtualBox, Linux, Git/GitHub

Frameworks & Libraries, Django/DRF, Flask, FastAPI, Laravel

LLMs & Agents, Multi-agent orchestration, RAG, prompt engineering

Extracurricular Activity

CTF Player

CTF COMPETITIONS CONSIST IN COMPETITIVE CHALLENGES THAT COVER VARIOUS TOPICS OF CYBERSECURITY

2023 - Present

- Competed in multiple national and international CTF competitions specializing in binary exploitation and reverse engineering.
- Ranked Top 1% globally on Root-Me (Rank: 1178).
- Authored technical write-ups detailing different types of security exploitation that can be found **here**.
- I participate with both teams *noreply* and *th3_r0n1ns*.

Cyber Security team leader

Alger, Algeria

GDG ALGIERS

Oct. 2023 – Jul. 2024

- Spearheaded and organized the club's cybersecurity initiatives and activities.